

Shuli Huang

Postdoctoral Scholar Research
Morsani College of Medicine, University of South Florida

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I am a cardiovascular physiologist with expertise in neuroendocrinology and immunology. My research uses rodent models to define neuroendocrine mechanisms underlying stress-responsive social behavior, focusing on oxytocinergic and vasopressinergic regulation of aggression and their cardiovascular and neuroendocrine consequences. In parallel, I investigate myocardial infarction-induced ischemic heart failure, examining how innate and adaptive immune cells regulate post-infarction cardiac repair and remodeling, with emphasis on T cell lymphopenia, age-related thymic involution, and innate immune memory.

EDUCATION

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|-------------|---|----------------|
| 2014 – 2021 | PhD Ecology
Institute of Zoology, University of Chinese Academy of Sciences
Supervisor: Dr. Zhibin Zhang | Beijing, China |
| 2010 – 2014 | B.S. Biological Science
Anhui Agricultural University | Hefei, China |

EMPLOYMENT

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|----------------|---|----------------------|
| 2025 - Present | Postdoctoral Scholar Research
University of South Florida
Supervisor: Dr. Yonggang Ma | Tampa, Florida |
| 2024 – 2025 | Postdoc Research Fellow
University of Mississippi Medical Center
Supervisor: Dr. Alex da Silva & Dr. Ji Li | Jackson, Mississippi |
| 2022 – 2024 | PostDoc
Hainan Seed Industry Laboratory
Supervisor: Dr. Zhibin Zhang & Dr. Jiaming Zhang | Sanya, China |

PUBLICATIONS

- Huang, S.**, Nunez, J., Toresco, DL., Wen, C., Slotabec, L., Wang, H., Zhang, H., Rouhi, N., Adenawoola, MI., Li, J. (2025). Alterations in the inflammatory homeostasis of aging-related cardiac dysfunction and Alzheimer's diseases. *FASEB Journal*, 39(1): e70303.
- Li, G., Zhang, M., **Huang, S.**, He, H., Wan, X., Wang, F., & Zhang, Z. (2025). Density-Dependent Difference in Serum Levels of Immunoglobulin G in Brandt's Voles and Its Potential Influencing Factors. *Integrative Zoology*, 20(5), 963-970.
- Zhang, X., Firman, R. C., Song, M., Li, G., Cheng, C., Liu, J., **Huang, S.**, Batsuren, E., Zhang, Z. (2023). Male and female Brandt's voles have higher reproductive success when they have more mating partners regardless of population density. *Behavioral Ecology*, 34(4), 662-672.

- Lu, W., **Huang, S.**, Liu, J., Batsuren, E., Li, G., Wan, X., Zhao, J., Wang, Z., Han, W., & Zhang, Z. (2023). Effects of group size on behavior, reproduction, and mRNA expression in brains of Brandt's Voles. *Brain Science*, 13(2).
- Zhao, J., Lu, W., **Huang, S.**, Le Maho, Y., Habold, C., Zhang, Z. (2022). Impacts of dietary protein and niacin deficiency on reproduction performance, body growth, and gut microbiota of female hamsters (*Tscherskia triton*) and their offspring. *Microbiology Spectrum*, 10(6), e0015722.
- Batsuren, E., Zhang, X., Song, M., Wan, X., Li, G., Liu, J., **Huang, S.**, Zhang, Z. (2022). Density-dependent changes of mating system and family structure in Brandt's voles (*Lasiopodomys brandtii*). *Ecology and Evolution*, 12(8), e9199.
- Liu, J., **Huang, S.**, Zhang, X., Li, G., Batsuren, E., Lu, W., Xu, X., He, C., Song, Y., Zhang, Z. (2021). Gut microbiota reflect the crowding stress of space shortage, physical and non-physical contact in Brandt's voles (*Lasiopodomys brandtii*). *Microbiological Research*, 255, 126928.
- Huang, S.**, Li, G., Pan, Y., Liu, J., Zhao, J., Zhang, X., Lu, W., Wan, X., Krebs, C. J., Wang, Z., Han, W., Zhang, Z. (2021). Population variation alters aggression-associated oxytocin and vasopressin expressions in brains of Brandt's voles in field conditions. *Frontiers in zoology*, 18(1), 56.
- Li, G., Wan, X., Yin, B., Wei, W., Hou, X., Zhang, X., Batsuren, E., Zhao, J., **Huang, S.**, Xu, X., Liu, J., Song, Y., Ozgul, A., Dickman, C. R., Wang, G., Krebs, C. J., Zhang, Z. (2021). Timing outweighs magnitude of rainfall in shaping population dynamics of a small mammal species in steppe grassland. *Proceedings of the National Academy of Sciences of the United States of America*, 118(42).
- Huang, S.**, Li, G., Pan, Y., Song, M., Zhao, J., Wan, X., Krebs, C. J., Wang, Z., Han, W., Zhang, Z. (2021). Density-induced social stress alters oxytocin and vasopressin activities in the brain of a small rodent species. *Integrative Zoology*, 16(2), 149-159.
- Liu, J., **Huang, S.**, Li, G., Zhao, J., Lu, W., Zhang, Z. (2020). High housing density increases stress hormone- or disease-associated fecal microbiota in male Brandt's voles (*Lasiopodomys brandtii*). *Hormones and Behavior*, 126, 104838.
- Zhao, J., Li, G., Lu, W., **Huang, S.**, Zhang, Z. (2020). Dominant and subordinate relationship formed by repeated social encounters alters gut microbiota in greater long-tailed hamsters. *Microbial Ecology*, 79(4), 998-1010.

RESEARCH SKILLS

Data Analysis & Programming

- R (statistical analysis and coding)
- GraphPad Prism (data analysis and visualization)

Animal Research, Behavioral Assays & Surgical Models

- Small mammal trapping and mark-recapture experiments
- Animal behavior tests: Open Field, Elevated Plus Maze, Barnes Maze
- Stereotaxic brain injection surgery
- Myocardial infarction surgery
- Thymus transplantation and thymectomy
- Echocardiography
- IonOptix contractility analysis

Molecular, Histological & Neuroscience Techniques

- Brain tissue sectioning

- Immunohistochemistry
- PCR and Western blotting

Certification

- Teacher Qualification License (Biology), China

REFEREES

Dr. Yonggang Ma (Postdoc Supervisor)

Assistant Professor, University of South Florida
yma4@usf.edu

Dr. Yvon Le Maho (Collaborator)

Professor, University of Strasbourg
yvon.lemaho@iphc.cnrs.fr

Dr. Xinru Wan (Lab colleague)

Professor, Jiangxi Agricultural University
wanxinru@jxau.edu.cn

Dr. Alexandre da Silva (Postdoc Supervisor)

Associate Professor, University of Mississippi Medical Center
asilva@umc.edu